

E-COMMERCE DATA ANALYSIS – BUSINESS INSIGHTS

Project Overview

This project analyzes the Olist E-commerce dataset using PostgreSQL. A total of 33 SQL queries were written to understand revenue trends, customer behavior, product trends, seller performance, and delivery efficiency.

Business Problem

E-commerce platforms generate large volumes of transactional data including orders, payments, products, customers, and delivery information. However, without proper analysis, this data cannot provide meaningful business insights.

Businesses need to understand:

- Which products generate the most revenue
- Which customers drive sales
- How sellers perform on the platform
- Whether delivery operations are efficient
- How customer purchasing behavior evolves over time

This project aims to answer these questions using SQL-based data analysis.

Dataset Description

The dataset used in this project is the Olist E-Commerce dataset.

The analysis includes the following key tables:

- customers
- orders
- order_items
- order_payments
- products

These tables capture the complete lifecycle of an e-commerce transaction including customer details, order information, product data, seller performance, and payment information.

DATASET OVERVIEW (EXPLORATORY ANALYSIS)

The initial exploratory queries were used to understand the overall scale of the dataset.

Total Customers: ~99,000

Total Orders: ~99,000

Total Products: ~33,000

Total Items Sold: ~112,000

Total Revenue Generated: ~16 million

These exploratory results provide a high-level overview of the platform's scale and business activity.

REVENUE INSIGHTS

1. Top Revenue Generating States

The state of São Paulo (SP) generated the highest revenue with approximately 5.99 million in sales.

Other top contributing states include Rio de Janeiro (RJ) and Minas Gerais (MG).

This indicates that the majority of the platform's revenue is concentrated in a few high-population regions.

2. Top Revenue Generating Product Categories

The category “cama_mesa_banho” generated the highest revenue (around 1.7 million), followed by:

- beleza_saude
- informatica_acessorios
- moveis_decoracao

These categories represent strong consumer demand in home and lifestyle products.

3. Revenue Growth Over Time

Revenue increased significantly over the years:

2016 → ~59K

2017 → ~7.24M

2018 → ~8.69M

This indicates rapid platform growth and increased customer adoption.

CUSTOMER INSIGHTS

1. Cities with the Highest Customers

São Paulo has the largest number of customers (15,540), followed by:

- Rio de Janeiro (6,882)
- Belo Horizonte (2,773)
- Brasília (2,131)

This shows that major metropolitan cities dominate the customer base.

2. Repeat Customers

Approximately 2,997 customers placed more than one order on the platform.

This indicates that while most customers purchase only once, there is still a significant segment of repeat buyers.

3. Average Orders per Customer

The average number of orders per customer is approximately 1 order per customer, indicating that most users are one-time buyers.

SELLER INSIGHTS

1. Average Revenue per Seller

The average revenue generated per seller is approximately 6,561.

This shows that seller revenue distribution varies significantly across the platform.

2. Top Sellers Contribution

A small group of sellers contributes a large share of total platform revenue.

This reflects the common e-commerce pattern where a small number of high-performing sellers dominate sales.

ORDER & DELIVERY INSIGHTS

1. Order Trends Over Time

Monthly order analysis shows steady growth in order volume as the platform expanded.

2. Average Delivery Time

The average delivery time between purchase and delivery provides insight into logistics efficiency and customer experience.

3. Late Deliveries

Approximately 8.11% of orders were delivered later than the estimated delivery date.

This suggests that the majority of deliveries are completed on time, but there is still room for improvement in logistics operations.

Business Recommendations

Based on the SQL analysis of the e-commerce dataset, the following business recommendations can help improve platform performance and customer experience:

1. Strengthen marketing efforts in high-revenue regions such as São Paulo and Rio de Janeiro, as these regions generate the largest share of platform revenue (with São Paulo alone contributing nearly 6 million). Targeted promotions and localized campaigns in these regions can further increase sales.
2. Focus on promoting high-performing product categories such as "cama_mesa_banho", "beleza_saude", and "informatica_acessorios", which currently generate the highest revenue. Highlighting these categories through platform recommendations, seasonal promotions, or bundled offers can drive additional growth.
3. Improve customer retention strategies since most customers place only one order and only about 2,997 customers are repeat buyers. Loyalty programs, personalized recommendations, and discount incentives for second purchases can help increase repeat purchase behavior.
4. Support and scale high-performing sellers, as the analysis shows that revenue distribution is uneven and a small group of sellers generates a large portion of platform sales. Providing these sellers with better visibility, marketing support, and operational assistance can further increase marketplace revenue.
5. Enhance logistics and delivery efficiency to reduce the current late delivery rate of approximately 8.11%. Improving delivery planning, optimizing warehouse locations, and strengthening logistics partnerships can further improve customer satisfaction.

CONCLUSION

The analysis reveals that revenue is concentrated in major cities and product categories, most customers place only one order, and logistics performance is generally reliable with a relatively low percentage of late deliveries.

These insights can help businesses improve customer retention strategies, optimize logistics, and focus on high-performing product categories.